DEEP LEARNING PROJECT PRESENTATION

AeroDepth-Lite

Monocular Drone-Camera Range Prediction and Adaptive Inference Scheduling

A compact depth model. A causal update policy.
An evidence-based study of accuracy and compute.

TEAM MEMBERS • EDIT THESE FIELDS

[Member name — Roll no.] | [Member name — Roll no.]

[Department] • [College / University]
Guide: [Name and designation]



Problem statement and motivation

How can a small drone estimate scene range while limiting repeated inference?

01 / PERCEPTION

One image, uncertain range

Recover dense metric range from monocular RGB under changing terrain, weather and visibility.

02 / COMPUTE

Repeated inference has a cost

A fresh model call for every frame adds work. Reusing an old map can increase staleness and miss scene changes.

03 / EVALUATION

Average error can hide near errors

Overall gains may coexist with worse errors below 10 m. Model calls, update delay and measured time all matter.

Project gap

Evaluate compact range prediction and causal scheduling together, including near-field regressions and real timing.

Source: docs/FINAL_REPORT.md · README.md

Objectives

Measure prediction quality, update behaviour and actual execution cost.

01	**Predict dense range**	Use a compact MobileNetV3-Small encoder and skip decoder to produce a 192 × 320 range map.
02	**Compare learning strategies**	Evaluate the original student, expanded supervised A and exploratory knowledge distillation (KD).
03	**Schedule causally**	Use current RGB, simulated IMU, held GPS and prior predictions to choose fresh inference or reuse.
04	**Verify the trade-offs**	Report overall / near error, model calls, update delay, measured timing and limitations.

Expected outputs

Reproducible model exports, frozen split manifests, comparative evidence and a local demonstration dashboard.

Source: docs/FINAL_REPORT.md · results/presentation_validation_20261029T155521Z/REPORT.md

Literature review and research gap

Method foundations and the limitations relevant to this project.

Study	Model / approach	Limit relative to our project scope
Eigen et al., 2014	Coarse-to-fine monocular depth	NYU / KITTI results do not establish UAV scheduler behaviour.
Howard et al., 2019	MobileNetV3; hardware-aware design	Efficient encoder; metric range needs a decoder and task-specific training.
He et al., 2016	Residual learning / ResNet	Classification backbone; larger capacity does not guarantee a suitable depth teacher.
Hinton et al., 2015	Teacher-to-student distillation	Knowledge transfer does not guarantee improved near-range regression.
Fonder & Van Droogenbroeck, 2019	MidAir synthetic UAV dataset	Rich simulation data; real-flight generalization must be tested separately.

Research gap addressed

Study the joint range-quality / update-cost trade-off on a fixed UAV subset; retain negative findings and per-path checks.

References: Eigen (2014), Howard (2019), He (2016), Hinton (2015), Fonder & Van Droogenbroeck (2019). Links in notes.

Novelty and team contributions

The contribution is the implemented combination and its controlled evaluation.

METHOD

Compact metric-range model

1,027,489 parameters; MobileNetV3-Small with a custom skip decoder.

SYSTEM

Causal multi-signal scheduler

Combines visual change, simulated motion, prior predicted risk and map age.

EXPERIMENT

Matched A / KD comparison

Same starting student state and training budget; a failed teacher gate stays visible.

REPRODUCIBILITY

Auditable delivery

Grouped splits, frozen artifacts, near-field metrics, ablations and measured timing.

Source: src/aerodepth_lite/ · docs/FINAL_REPORT.md · saved student and scheduler comparisons

Dataset details

MidAir · synthetic multi-modal data for low-altitude drone flights

Project split	Frames	Paths	Status
Expanded train	44,098	12	Completed
Validation	4,410	2	Selection / analysis
Reserved new test	8,820*	4	NOT RUN

48,508 available train + validation pairs

Train: Kite 17,640 · PLE 26,458
Validation: Kite 2,205 · PLE 2,205
Classes / per-class counts: not applicable (dense regression).

*Expected from metadata; 8 RGB/depth archives missing. Legacy 4,436-frame test is historical and excluded from this split.

FORMAT & DATA QUALITY

RGB: 1024 × 1024 JPEG
Range: uint16 PNG / float16 bits
Sensors: HDF5 (IMU / GPS)

0 duplicate RGB paths in manifest.
Invalid range pixels are masked.
Near-field evaluation is imbalanced: 97.23% of near pixels are from PLE.

Dataset source / download

Sources: expanded_preparation_20261026/{manifest_train_val.json.gz,data_validation.json} · MidAir paper and FAQ

Data preprocessing and leakage controls

Preserve the range target and keep related flight paths in one split.

01 **Validate pairs** — Match RGB, range and sensor records using the frozen allowlist.

02 **Decode & mask** — Reinterpret PNG bits as float16 range. Keep finite GT in [0.1, 1000] m.

03 **Resize & normalize** — Full-frame 192 × 320; bilinear RGB; nearest target / mask; ImageNet normalization once.

04 **Augment training** — Paired horizontal flip for RGB, target and mask; no validation augmentation.

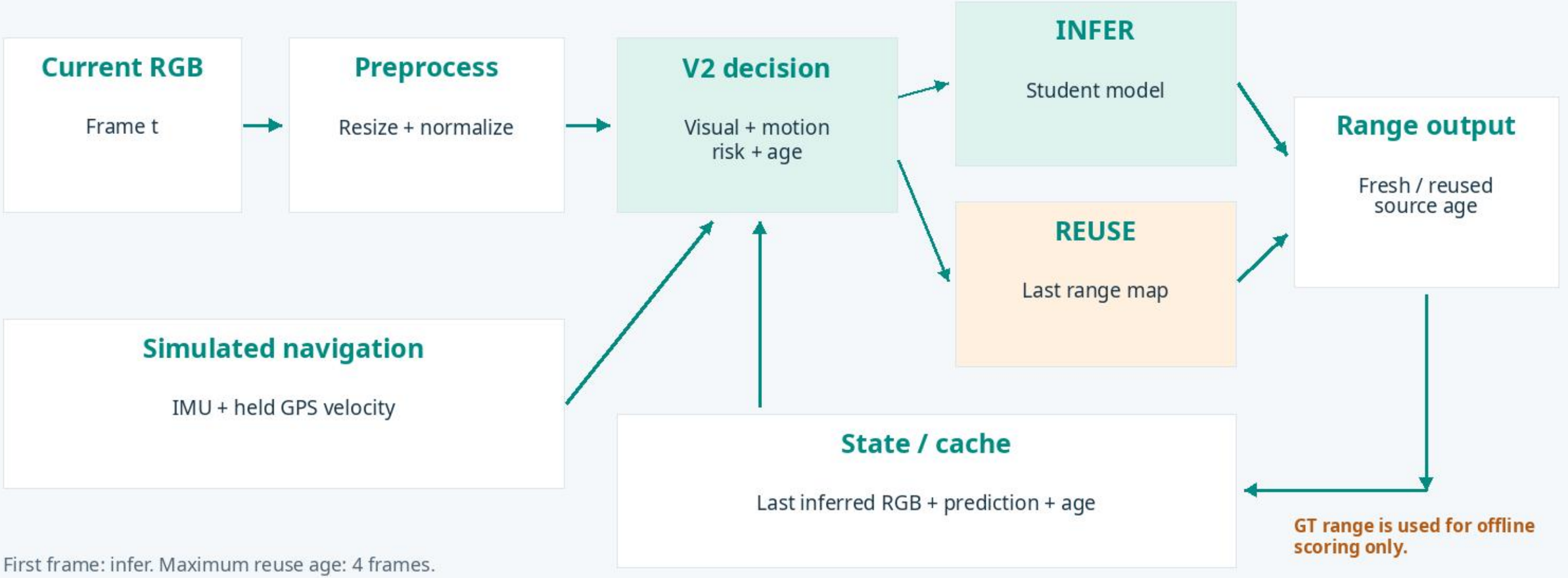
Leakage controls

Group weather variants of a path; quarantine legacy tests; select on validation only; keep current GT out of scheduler decisions.

Source: docs/expanded_training.md · expanded_preparation_20261026/dataset_contract.json · src/aerodepth_lite/data.py

Proposed system architecture

Causal runtime workflow · drawn with editable shapes and connectors



First frame: infer. Maximum reuse age: 4 frames.

Source: src/aerodepth_lite/{model.py,scheduler_v2.py,presentation_scheduler.py} · docs/FINAL_REPORT.md

Deep learning model and methodology

Small student for inference; larger teacher evaluated only as a training aid.

RGB → MobileNetV3-Small → 576→128 projection → skip decoder → 1-channel range

STUDENT ARCHITECTURE

1,027,489 parameters
Skip indices 0 / 1 / 3 / 8
Decoder widths 64 / 32 / 24 / 16
Bilinear upsampling; softplus + 0.1 m

Why selected: compact ImageNet-initialized encoder with a custom dense decoder. Teacher: ResNet-50 V1.5.

TRAINING CONFIGURATION

A / KD: 10 epochs each; selected epoch 9
Teacher B: 20 epochs; selected epoch 3
AdamW · batch 2 · seed 42 · clip 1.0
Student LR 10^{-4} · weight decay 10^{-4}
Cosine minimum: 10^{-6}
Teacher encoder / decoder LR: 10^{-5} / 10^{-4}

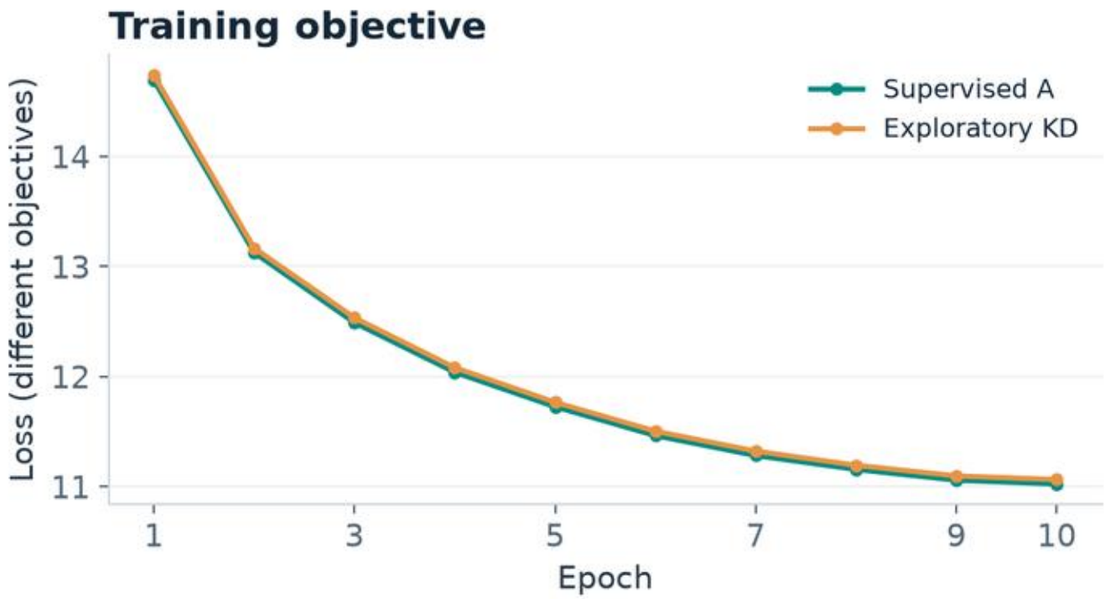
Tools: Python · PyTorch / torchvision · NumPy · Streamlit

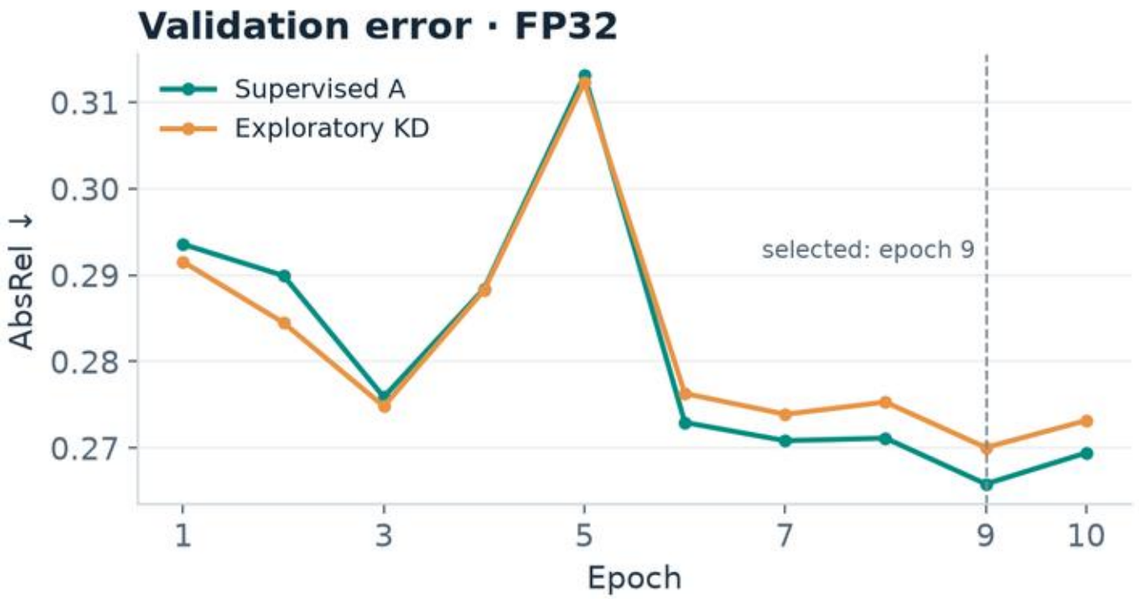
Loss = L1 range + 0.2 L1 log-range + 0.1 gradient L1 | KD adds 0.2 log-range consistency

Source: docs/FINAL_REPORT.md · frozen launch configurations · src/aerodepth_lite/model.py

Experimental results: training and validation

Recorded curves from the completed expanded supervised and exploratory KD runs.





Both students selected epoch 9

Selection metric: pooled FP32 validation AbsRel.

Lower training loss is not proof of generalization

A and KD objectives differ; compare validation error.

Source: results/presentation_validation_20261029T155521Z/training_curves.json · plots regenerated from saved values

Experimental results: model comparison

4,410 validation frames · lower error is better · identical evaluation masks

Model	AbsRel ↓	RMSE (m) ↓	δ1 ↑	Near RMSE ↓
Original · epoch 4	0.301540	127.335	49.80%	3.362
Supervised A · epoch 9	0.265831	119.944	62.62%	3.906
Exploratory KD · epoch 9	0.270049	121.207	61.81%	3.961

Teacher B failed its acceptance gate: Kite near RMSE 6.636 m vs A 4.695 m.

11.84%

Lower AbsRel · A vs original

16.18%

Higher near RMSE · A vs original

1.59%

Higher AbsRel · KD vs A

δ1 = fraction with max(prediction / GT, GT / prediction) < 1.25. Classification precision, recall, F1 and confusion matrix do not apply.

Source: results/student_validation_comparison_20261029T152648Z/comparison.csv · docs/FINAL_REPORT.md

Experimental results: scheduling and measured cost

Supervised A on validation · policy parameters frozen before this comparison

Policy	Calls	Near RMSE (m)	Max delay[1]
Every frame	4,410	3.906	0
Fixed budget	2,456	4.161	1
V2 adaptive	2,456	3.780	2
V2: no visual	1,789	5.529	4

9.14%

Lower pooled near RMSE than fixed budget at equal total calls.

Overall error slightly worsens; max event-update delay rises.

Actual timing · 256-frame stress windows · median end-to-end seconds

Device	Every frame	Fixed budget	V2 cached
CPU	10.420	8.673	10.429
GTX 1650	9.018	7.863	8.765

Risk caching reduced scheduler time by 13.43% on CPU and 15.81% on GPU.
No robust end-to-end speedup was measured.

Source: presentation_validation_20261029T155521Z/{scheduler_comparison.csv,timing_individual.csv} · docs/FINAL_REPORT.md

Limitations, future scope and conclusion

Overall validation gains are useful only when the near-field and compute trade-offs stay visible.

CURRENT LIMITATIONS

Evidence remains bounded

Two familiar validation paths; correlated frames; synthetic scenes; one seed; substantial depth error; near-field regression.

FUTURE SCOPE

Complete independent evaluation

Acquire the 8 missing reserved streams, freeze the test protocol, evaluate new paths, then study near-field losses and real-flight transfer.

CONCLUSION

Supervised A improves overall range prediction, while near-field error worsens.
Exploratory KD underperforms A; adaptive scheduling offers a mixed quality / cost trade-off.

Source: docs/FINAL_REPORT.md · results/acquisition_publication_20261029T180214Z/REPORT.md